

AEGISSHIELD INSURANCE LTD.

Insurance Business Analysis Project Report

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A practical data analytics project covering business performance, risk, retention and growth.

Tools & Technologies

PostgreSQL • Microsoft Excel • Power Query • Power BI

Data Analytics Portfolio Project

1. Executive Summary

This project analyzes the business performance of AegisShield Insurance Ltd. using PostgreSQL, Microsoft Excel, Power Query and Power BI. The analysis looks at customers, policies and premiums, claims, fraud risk, renewals, agents and customer complaints.

The final Power BI dashboard turns the analysis into a clear management view of business performance, risk, customer retention and growth. The report covers 100,000 policies, about ■6.11 billion in total premium, 22,000 claims, about ■674.41 million in claim amount, and an overall renewal rate of 78%.

The analysis also shows that Commercial generates the highest premium, Motor has the largest policy and claim volume, and Fire has the highest claim amount by claim type.

2. Project Objectives

- Understand customer distribution by segment, state, income band and age.
- Measure policy volume, premium contribution, average premium and sum insured.
- Analyze claims by status, claim type, product and monthly trend.
- Identify high-risk claims and understand fraud-risk patterns.
- Measure renewal performance by product, channel and agent.
- Evaluate agent productivity, premium contribution and retention performance.
- Analyze complaint volume, complaint channels and resolution performance.
- Turn the findings into practical business insights through an interactive Power BI dashboard.

3. Technology Stack & Role of Each Tool

Tool	How it was used
PostgreSQL	Data storage, SQL querying, filtering, aggregation and data preparation.
Microsoft Excel	Business analysis, validation, structured analysis and supporting calculations.
Power Query	Data extraction, cleaning, transformation, type conversion and reporting preparation.
Power BI	Data modeling, DAX measures, interactive dashboards, filters and business insights.

4. Data Preparation & Analytical Workflow

- The project follows a practical end-to-end analytics workflow:
 1. Insurance data is organized and queried in PostgreSQL.
 2. SQL is used to inspect records, filter relevant fields and create business-level summaries where required.
 3. The required data is brought into Excel and Power Query.
 4. Power Query is used for cleaning and transformation, including data types, dates, categories and reporting-ready fields.
 5. The prepared data is loaded into Power BI and the required relationships are created.
 6. DAX measures are created for KPIs such as Total Customers, Total Policies, Total Premium, Total Claims, Total Claim Amount and Renewal Rate.

7. Interactive report pages and slicers are created for product, state, channel, renewal status and other business dimensions.

8. Dashboard results are translated into business findings, impact and practical recommendations.

5. Dashboard Structure

Page	Main Focus	Purpose
01	Executive Overview	KPIs, product premium, policy distribution, premium trend and renewal rate.
02	Customer Analysis	Customer segment, state, income band, age and customer premium.
03	Policy & Premium Analysis	Product and channel performance, premium trends and portfolio summary.
04	Claims Analysis	Claim status, claim type, product exposure, trends and settlement time.
05	Fraud & Risk Analysis	Fraud-risk distribution, high-risk claim amount, product/type/provider risk and trends.
06	Renewal & Retention Analysis	Renewed vs non-renewed policies, product/channel renewal rates and agent retention.
07	Agent Performance Analysis	Agent premium, policy volume, renewal rate, branch contribution and agent-level details.
08	Complaint Analysis	Complaint type, channel, resolution status, resolution time and trends.
09–10	Business Insights	Detailed findings, evidence, business impact and recommended actions.

6. Key Business Findings

6.1 Executive Overview

- Commercial generates the highest premium at approximately **₹2.36 billion**.
- Motor has the largest policy share at about 38%, followed by Health at about 28%.
- Overall renewal rate is approximately 78%.
- Fire and Damage have the highest claim amounts, at approximately **₹99.66 million** and **₹98.89 million** respectively.
- Monthly premium varies over the displayed period, with the highest level around August 2025 at approximately **₹272 million** and the lowest around February 2025 at approximately **₹225 million**.

6.2 Customer Analysis

- Retail is the largest customer segment with 62,083 customers, followed by Affluent with 19,887.
- Maharashtra has the highest customer concentration with 15,002 customers.
- Retail contributes approximately **₹3.79 billion** in premium, making it the largest customer-segment premium contribution.
- The dashboard also provides customer analysis by segment, state, income band and age.

6.3 Policy & Premium Analysis

- The portfolio contains 100,000 policies and approximately **₹6.11 billion** in premium.

- Average premium is approximately ■61.06K and total sum insured is approximately ■284.11 billion.
- Commercial contributes approximately ■2.36 billion in premium, the highest among products.
- Agent is the largest premium channel at approximately ■1.94 billion.
- The monthly premium shown in the dashboard ranges from approximately ■225 million to ■272 million.

6.4 Claims Analysis

- There are approximately 22K claims with a total claim amount of ■674.41M.
- Total approved amount is approximately ■399.68M and average claim amount is ■30.37K.
- Motor has the highest claim volume at approximately 8.36K and the highest product claim amount at approximately ■256.13M.
- Fire has the highest claim-type amount at approximately ■99.66M.
- Average settlement time is approximately 38.55 days overall, with several claim types around 39 days.
- 15,935 claims are settled, 2,858 are pending, 2,261 are rejected and 1,154 are under investigation.

6.5 Fraud & Risk Analysis

- 220 claims are classified as high risk, which is approximately 1% of total claims.
- High-risk claim amount is approximately ■7.19M.
- Motor has the highest high-risk claim amount at approximately ■2.82M.
- Theft has the highest high-risk claim amount by claim type at approximately ■1.48M.
- Local Provider records the highest displayed average fraud score at approximately 22.

6.6 Renewal & Retention Analysis

- 78,168 of 100,000 policies were renewed, giving a renewal rate of 78.17%.
- 21,832 policies were not renewed, representing 21.83%.
- Personal Accident and Health show approximately 79% renewal, while the other displayed products are approximately 78%.
- Renewed policies contribute approximately ■4.77 billion in premium compared with approximately ■1.34 billion from non-renewed policies.
- The displayed high-performing agents have renewal rates around 81%–84%.

6.7 Agent Performance Analysis

- The dashboard covers 300 agents managing approximately 100K policies.
- Agent_5095 contributes the highest displayed premium at approximately ■26.35M.
- Agent_5280 has the highest displayed policy volume at approximately 383 policies.
- Agent_5139 has the highest displayed renewal rate at approximately 84%.
- Central branch contributes the highest premium at approximately ■1.54 billion.
- Overall agent-network premium is approximately ■6.11 billion with a 78% renewal rate.

6.8 Complaint Analysis

- Approximately 12K complaints were recorded.
- 11K complaints were resolved and approximately 1K remained open.
- Complaint resolution rate is 89.59%, with an average resolution time of 15.52 days.
- Claim Delay is the most common complaint type with 1,759 complaints, followed by Documentation with 1,730.
- Agent has the highest complaint volume at 2,039, closely followed by Call Centre at 2,030.
- Policy Cancellation has the longest average resolution time at approximately 15.8 days.

7. Business Recommendations

- Protect and retain the Commercial portfolio because it generates the highest premium, while continuing to monitor its claims and complaint performance.
- Use customer segment and geographic concentration information to plan targeted retention, acquisition and cross-selling activities.
- Review Motor portfolio risk and claims patterns because Motor has the highest policy and claim volume.
- Investigate major Fire claim patterns and strengthen risk and claims monitoring.
- Prioritize high-risk Motor and Theft claims for investigation and review recurring risk patterns.
- Improve retention in lower-performing products or channels by studying the practices of higher-retention groups.
- Review Accident claims settlement processes where settlement time is around 39 days.
- Reduce Claim Delay complaints and work on the open complaint backlog.
- Review Policy Cancellation processes because they have the longest complaint-resolution time.
- Compare high-premium agents with high-renewal agents to identify practices that can be shared across the network.

8. Conclusion

This project demonstrates an end-to-end insurance analytics workflow using PostgreSQL, Excel, Power Query and Power BI. The analysis brings together customer, policy, premium, claims, fraud risk, renewal, agent and complaint information in one reporting solution.

The final dashboard provides measurable KPIs and supporting business insights that can help management monitor business performance, risk exposure, customer retention and service quality. It also shows how raw insurance data can be turned into a clear and useful business story.

9. Project Deliverables

- PostgreSQL-based insurance data analysis and querying.
- Excel business-analysis workbook and analysis log.
- Power Query data-cleaning and transformation workflow.
- Power BI interactive insurance business dashboard.
- Executive-level business insights and recommendations.
- Final project report documenting the objectives, tools, analysis and findings.